GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. We prove a stopped acquisition–memory theorem from raw causal experiments by balancing their report measures and unnormalized task-loss measures together. A single admissible reference experiment supplies the converse against every history-dependent procedure, while one parameter-independent finite-state program supplies the upper for all hidden dynamics. With exact task balance and compatible physical ports, the minimax risk is exactly a density-constrained quantization profile of a genuinely acquired law, at the paid productive budget; all live and terminal states are charged. Finite report transformations give a controlled terminal-alphabet expansion, rather than a free decoder seed. The result does not require a positive lower bound on every conditional variance, a common primary row under every strategy, or a contracting normalized filter. A raw moment criterion characterizes finite balancing mixtures, and an integrated instrument estimate controls rank-changing updates without division by small report probabilities. Controlled classical kernels and noncommuting quantum instruments verify the same theorem, including zero positivity floors and singular acquired supports. A nonintervening Gaussian sensing problem has an exact one-value obstruction and a uniform, asymptotically sharp two-value waiting law. Relative acceptance certificates and typed causal simulation complete the resource and error composition. The inherited marked covering calculation is reproduced with its actual mass factors; it is not a new quantization formula.

1. Introduction

A rare favorable report has no operational meaning until the experiment that obtains it has been paid for. A predictive state has no finite-memory meaning until its update, legal commands, stopping decision and task readout have been implemented. These two observations lead to the same question: which geometry of an actually prepared causal experiment controls its attainable prediction risk under simultaneous acquisition and memory constraints?

There is a further obstruction when the hidden dynamics are unknown. A lower bound on conditional audit variance can fail after a highly informative history, even though a single program remains uniformly minimax. Imposing a positive variance floor on every history can therefore confuse a sufficient pointwise calibration mechanism with the resource law itself. This paper separates the two. The quantities to balance are the report measure and the *unnormalized* squared-loss measure. Balancing the first without the second does not survive report-dependent selection.

The central result, Theorem 3.1, starts with positive prepared kernels, a declared scored future test, paid calls and a physical stopping interface. It derives a density-limited acquired measure from a fixed admissible reference model. It then constructs one common rejection program from a task-balancing identity certified before choosing the memory budget. In the exact case the checkpoint risk is $\mathcal{Q}_m(N/\kappa)$ and the autonomous risk is $\mathcal{Q}_{M-B}(N/\kappa)$, where B is a proved number of mandatory live states. The reference model is a genuine member of the uncertainty class, not an auxiliary distribution selected after seeing the code. Conversely, the implemented upper does not know that model. Its full hidden state may remain nonstationary and nonmixing.

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The theorem also permits a finite family of report and readout transformations. Their exported terminal dictionary is counted: m virtual values may require Dm physical values. Under acquired covering and small-ball hypotheses this yields matching resource orders; in invariant examples the alphabet expansion can be avoided. The predictive quotient is constructed only after specifying executable tests and measurable realization conditions. A task statistic is not silently promoted to the full predictive quotient.

The principal additions are the common raw task-measure criterion, its exact stopped minimax completion and its verification without a positivity floor. Proposition 5.1 gives a necessary and sufficient linear criterion for a specified finite family of balancing commands; it is not a characterization of all possible causal observers. Proposition 5.2 proves the integrated continuity needed at singular reports. Theorem 7.2 and Theorem 7.3 verify the central theorem by genuinely different raw instruments. Theorem 8.1 strengthens the retained Gaussian application [1] to a zero-prior-floor uncertainty class and a ratio-one risk asymptotic: one terminal value cannot exploit waiting in the worst case, whereas two values achieve the asymptotically optimal waiting law, uniformly over arbitrary additional memory.

The underlying tools have clear antecedents. Randomization over a finite symmetry group is standard invariant decision theory [3]; Pauli randomization is classical quantum information [4]. Neither is claimed as a new operation. Optimal rejection rules go back to Chow [5]; thresholding conditional variance in square-loss regression with a fixed rejection rate is established in [6]. Our Gaussian calculation does not claim either principle. Its additional assertions concern predictable paid waiting, a common rule over history-dependent hidden dynamics, the exact one-value obstruction and an asymptotic sharpness statement uniform over all larger finite terminal alphabets. Occupation-measure Markovization and absorbing control are developed in [7, 8, 9, 10]. The present proof does not identify all hidden histories with a known finite driver or infer compactness of a general occupation set. It uses predictable counts under one witness and history-uniform task-measure identities for one common upper program. General finite-memory POMDP reachability remains a different and computationally difficult problem [11, 12]. The marked reliability–resolution calculation and physical-port accounting were developed in the retained causal-acquisition and predictable-geometry manuscripts [1, 2]; their proofs and original scope are retained and the needed arguments are repeated here.

The limits are also mathematical. Existence of a balancing command and of a suitable reference member are substantive assumptions. No universal learning rate for unknown kernels, no unrestricted equivalence of interventions on different targets, and no efficient synthesis theorem is asserted. The exact profile is first a theorem for Borel finite-state transducers. Proposition 9.2 supplies finite-description implementations under explicit effective report and probability certificates, with their precision and workspace charged separately. This distinction is necessary even for ordinary continuous-observation experiments.

2. Prepared experiments and balanced task measures

All state, action and report spaces are standard Borel. An unknown index $\theta \in \Theta$ specifies a prepared law and positive normalized causal kernels. It is fixed for an entire execution; it may index an arbitrary history-dependent physical evolution. Policies see only past reports, legal physical commands and their own allowed coins. They never see θ. All assertions below are for the declared experiment and test, not for a formal parameter manifold.

A call has a nonnegative cost bounded below by $c_* > 0$. A predictable flag declares whether the call is productive. Commitment is allowed only after a productive report. Conditional on commitment, a fixed bounded future audit $Z \in H$ is executed, where $H \subset \mathbb{R}^k$ is compact and convex. This audit is not observed before the prediction is committed. The loss is $\|Z - r\|^2$, $r \in H$. Each productive call has a potential audit specified by the same kernel, whether or not the observer commits there. A fresh stopping coin is independent of that audit conditional on the current physical

report and history. All other informative components of the current report must be included in the conditional laws below.

The cost N in the core theorem counts preparation, failed gate calls, productive calls, and all calls preceding commitment. A separately required final physical audit has its own explicit cost. Thus a protocol with one initial preparation call and one final audit uses total raw budget $N+2$, not N. Time, bit operations and scratch space are not identified with raw-call count.

A checkpoint procedure may retain the full precommit history, but its committed value belongs to a single fixed dictionary of at most m elements. An autonomous observer has a fixed alphabet of at most M states, including every live routing state and every terminal value. Its command law and readout are determined by its state and its current permitted input. At the interface between calls its only observer memory is that state. Its fixed program and real constants are part of the ideal Borel model; finite description is treated separately. Fresh coins can be used within a call but no seed, clock, chosen dictionary or subprogram persists for free. In particular, randomly selecting a different read-only program for the whole execution is not an uncharged random policy. With robust almost-sure commitment and robust expected cost at most N, write the infimal worst-case risks as $R^{\Theta}_{\mathrm{cp}}(N,m)$ and $R^{\Theta}_{\mathrm{aut}}(N,M)$; the infimum over an empty class is $+\infty$.

A genuinely acquired reference. Fix one admissible $\theta_0 \in \Theta$ and a known, explicitly certified probability kernel $K_0(dx,dz)$ on $X \times H$. Under θ_0, after *every* legal productive command and every surviving pre-call history, the conditional mark law is $\nu(dx) = K_0(dx,H)$. Conditional on the full current report, the audit has the same first and second moments as $K_0(dz \mid x)$. Known within-call command coins or auxiliary reports supply no further audit information under this witness. Define, ν-almost everywhere,

$$f(x) = \int z K_0(dz \mid x), \qquad w(x) = \int \|z - f(x)\|^2 K_0(dz \mid x), \qquad d_r(x) = w(x) + \|f(x) - r\|^2. \tag{2.1}$$

Take Borel representatives, with $f \in H$, $w \geq 0$. Thus the geometry is derived from an executed report–audit law. It is not a covariance or a reference volume appended independently to the experiment.

Assume that every legal observer satisfies, under this *same* witness,

$$\mathbb{E}_{\theta_0} C_\tau \geq \kappa \mathbb{E}_{\theta_0} A_\tau, \qquad A_\tau = \text{number of productive calls through commitment}, \tag{2.2}$$

where $\kappa > 0$ and C_τ is its actual accumulated core cost. This is an experiment-level all-strategy inequality. It may follow from a predictable compensated count or a bounded potential, and will be proved directly in the realizations. Since $A_\tau \geq 1$, $N < \kappa$ is infeasible under the witness.

One common upper program. Fix a parameter-independent acquisition program $\mathcal{G}$ using B live states. From its start state it obtains one productive report, almost surely, at conditional expected cost at most κ on *every* surviving history and every θ. After rejecting that report it returns to the same program start. This is a reset of the observer's routing label, not an assumed physical reset of hidden dynamics.

Within the productive call, $\mathcal{G}$ is allowed a fresh finite choice $j \in J$, a Borel report map to a virtual mark x, and a Borel export map $T_j : H \to H$. No information about θ is used. For every pre-attempt history h, every θ, every Borel $D \subset X$ and every $r \in H$, require

$$\mathbb{P}_\theta(x \in D \mid h, \mathcal{G}) = \nu(D), \tag{2.3}$$

$$\mathbb{E}_\theta[\mathbf{1}_{\{x \in D\}} \|Z - T_j r\|^2 \mid h, \mathcal{G}] = \int_D d_r(x) \nu(dx). \tag{2.4}$$

The physical reference mark used in the lower and the virtual mark used in the upper may be different report functions; they take values in the same declared space and their compared law is the same actually prepared ν.

The averaging in these identities includes the fresh choice j, gate history inside the attempt, and all hidden states, but the rule later applied to the virtual mark may not retain the averaging coin except through the exported terminal value. The identities are verified from the raw kernels *before* selecting m, N or a dictionary. They neither require equality of the full hidden law nor assert equal conditional variances after conditioning on j.

There is a declared export multiplicity $D_0 \in \mathbb{N}$ such that

$$|\{T_j r : r \in C,\ j \in J\}| \le D_0|C| \quad \text{for every finite } C \subset H. \tag{2.5}$$

The important exact case is $T_j r = r$, $D_0 = 1$. Finite isometry groups give $D_0 \le |J|$, but are not required by the theorem. All routing needed to perform these operations between calls is already included in B. A random pulse performed in a separate call needs its own retained control state and cost.

The loss identity is an identity of unnormalized measures, not of posterior variances after averaging a normalized filter. It is equivalent to the corresponding order-zero, first and squared-norm moment identities when T_j is an isometry, as made precise in Proposition 5.1. It allows arbitrary report dependence and zero residual variance on individual actual histories.

3. The acquired-geometry transfer theorem

For $b \ge 1$ and integer $m \ge 1$, set

$$\mathcal{Q}_m(b) = \min_{\substack{C \subset H,\ 1 \le |C| \le m \\ 0 \le h \le b,\ \int h\, d\nu = 1}} \int [w(x) + \operatorname{dist}(f(x), C)^2] h(x) \nu(dx). \tag{3.1}$$

Set $\mathcal{Q}_m(b) = +\infty$ when $b < 1$ or $m < 1$. The density cap is relative to the actually acquired ν. It does not condition on a favorable event without retaining its mass.

Theorem 3.1 (Balanced causal acquisition and finite-state completion)**.** *Suppose the raw conditions of Section 2 hold, with the reference and effort inequality using the same fixed admissible model. Let $b = N/\kappa \ge 1$. Then*

$$\mathcal{Q}_m(b) \le R^{\Theta}_{\mathrm{cp}}(N, m), \tag{3.2}$$

$$R^{\Theta}_{\mathrm{cp}}(N, D_0 m) \le \mathcal{Q}_m(b), \qquad R^{\Theta}_{\mathrm{aut}}(N, B + D_0 m) \le \mathcal{Q}_m(b), \tag{3.3}$$

$$\mathcal{Q}_{M-1}(b) \le R^{\Theta}_{\mathrm{aut}}(N, M). \tag{3.4}$$

If, under that witness, every terminating legal observer must use at least B live states in the physical sense of Proposition 4.4, replace $M - 1$ in (3.4) *by $M - B$. In particular, when $D_0 = 1$ and this lower is certified, for every $N \ge \kappa$ and $m \ge 1$,*

$$R^{\Theta}_{\mathrm{cp}}(N, m) = \mathcal{Q}_m(N/\kappa), \qquad R^{\Theta}_{\mathrm{aut}}(N, B + m) = \mathcal{Q}_m(N/\kappa). \tag{3.5}$$

The infima in these equalities are attained in the declared Borel model. For $N < \kappa$, no legal robust observer exists. The autonomous class is also empty for $M \le B$ when the mandatory-state certificate applies.

This is a minimax theorem with one common upper program, not a family of modelwise reductions. The two ingredients are logically different: the reference quantifies over *all* competing strategies at one actual model, while balance quantifies over *all* models and histories for one common program. The latter need not be a full predictive-state simulator. For nontrivial export multiplicity, an invariant optimal dictionary C satisfying $\bigcup_j T_j C = C$ gives the corresponding upper with $|C|$, not $D_0|C|$, terminal values.

Corollary 3.2 (Regular resource orders with exported values)**.** *Suppose the actual marked geometry satisfies the hypotheses of Proposition 6.1 with $d, \alpha > 0$, and the mandatory live-state count is*

B. For fixed D_0 and $M - B \geq D_0$, both bounds in Theorem 3.1 are comparable, with constants independent of N, M, to

$$F(N/\kappa, M - B) = \max\left\{(N/\kappa)^{-2/\alpha}, ((N/\kappa)(M - B))^{-2/(d+\alpha)}\right\}. \tag{3.6}$$

For checkpoint risk use m in place of $M - B$, with $m \geq D_0$. The constants depend on the stated geometry and D_0, not on an ambient parameter dimension or on a hidden positivity floor.

Proof. Apply Proposition 6.1 to the lower width $M - B$ and the upper virtual width $\lfloor (M - B)/D_0 \rfloor$. This integer is at least $(M - B)/(2D_0)$ in the declared range. The displayed function changes by at most a fixed factor under that width change. The checkpoint argument is identical. □

Atoms must be handled using (3.1), not by replacing it with a power law. Proposition 6.2 preserves their exact saturation threshold when $D_0 = 1$. Neither the lower nor the upper calls a conditional acquired law a free prepared sample.

4. Proof of the transfer and physical memory bounds

Lemma 4.1 (Attainment and continuity of the acquired profile)**.** *The minimum in* (3.1) *is attained. For each fixed m, $\mathcal{Q}_m$ is nonincreasing and left-continuous on $(1, \infty)$.*

Proof. Represent a dictionary by an ordered m-tuple in the compact space H^m, permitting repetitions. The feasible densities are a weak-star compact subset of the radius-b ball of $L^\infty(\nu)$: positivity, the upper bound and the mass equality are weak-star closed. Nets suffice; no unmentioned sequential compactness is used. The bounded loss functions vary uniformly continuously with the tuple, since squared distance on H is uniformly Lipschitz in its second argument. Along a convergent product net, split the objective difference into this uniform difference and the weak-star pairing with one fixed bounded measurable loss. This proves attainment. Monotonicity is immediate. If h is optimal at $b > 1$, use $h_\lambda = (1 - \lambda)h + \lambda$, whose cap is $b_\lambda = (1 - \lambda)b + \lambda < b$. Its loss tends to the optimal loss as $\lambda \downarrow 0$. Monotonicity sandwiches the values and proves the last assertion. □

Lemma 4.2 (A terminal measure obtained by any competitor)**.** *Under θ_0, every legal terminating procedure of expected cost at most N has a terminal mark measure ζ satisfying*

$$\zeta(X) = 1, \qquad 0 \leq \zeta \leq (\mathbb{E}_{\theta_0} A_\tau)\nu \leq (N/\kappa)\nu. \tag{4.1}$$

If its terminal dictionary is C, its expected loss is at least $\int d_C \, d\zeta$, where $d_C = w + \operatorname{dist}(f, C)^2$.

Proof. For a Borel set D, predictability of productive calls and the conditional reference row imply, by Tonelli,

$$\mathbb{E}_{\theta_0} \sum_{t \leq \tau} \mathbf{1}_{\{t \text{ productive}\}} \mathbf{1}_{\{X_t \in D\}} = \mathbb{E}_{\theta_0} A_\tau \, \nu(D).$$

The terminal report is one of these reports, so its measure is dominated by this acquired report-count measure. Almost-sure commitment gives mass one, and (2.2) gives the last inequality. Only now take its Radon–Nikodym density, choosing a Borel version clipped to the displayed cap.

Condition on the entire current report, the preceding history and the chosen output. Under the witness, the conditional squared loss is $w(X) + \|f(X) - r\|^2$. The fresh stopping coin leaves this identity unchanged. Randomizing among a fixed dictionary cannot reduce the value below its nearest point. Summing over the disjoint commitment events proves the claim. No independence between different calls was assumed. □

Lemma 4.3 (One balanced rejection program)**.** *Let (C, h) be feasible in* (3.1)*. After each attempt of $\mathcal{G}$, accept virtual mark x with probability $h(x)/b$ and export $T_j r_C(x)$, where r_C is the first nearest point to $f(x)$ in a fixed ordering of C. Otherwise return to the start label. This program has expected core cost at most κb, risk exactly $\int d_C h \, d\nu$ under every θ, and at most $B + D_0|C|$ states.*

Proof. Borel representatives and the finite nearest-point rule make the program measurable. On every surviving pre-attempt history, (2.3) gives acceptance probability $1/b$. For each nearest-point cell, apply (2.4) with its constant center, first to simple acceptance functions and then by bounded approximation to h/b. The expected terminal loss contributed by that attempt is

$$b^{-1}\int d_C(x)h(x)\nu(dx).$$

Consequently the probability of surviving j complete attempts is $(1-1/b)^j$. This is a consequence of the conditional hazard, not an assumed independent renewal structure. Summing the predictable attempt costs gives $\mathbb{E}C \le \kappa\sum_{j\ge 0}(1-1/b)^j = \kappa b$. Summing the terminal contributions gives the stated risk. For $b=1$ the first attempt is accepted almost surely. The physical state need not return to its former distribution on rejection. Only the B routing labels and the exported terminal dictionary persist. The current choice j is discarded after its effect has been encoded in that terminal value or the routing label. Thus the state count follows from (2.5). □

Proposition 4.4 (Semantic, presentation-invariant live-state bound)**.** *Consider finitely many mandatory pre-call contexts I with no additional free context input to the observer. A physical command includes its legal histories, executed kernel, port, visible reports and costs, not merely its printed name. Assume that the legal physical command set at every history in context i is exactly $\mathcal{U}_i$. Suppose that, under one fixed admitted model, every legal terminating procedure visits each context with positive probability. Let b_{phys} be the least number of subsets covering I such that each subset J has $\bigcap_{i\in J}\mathcal{U}_i \neq \varnothing$. Every autonomous observer needs at least b_{phys} live states. This count is invariant under duplicating command names or bijective stateless recoding of the same physical commands.*

Proof. For a live label that occurs in a collection J of mandatory contexts, its fixed command distribution must be supported on commands legal in every member of J. Null exceptional commands can be removed simultaneously because J is finite. Thus the intersection is nonempty. The collections served by its live labels cover the mandatory contexts, proving the bound. Duplicating a name changes neither the physical intersections nor this covering number; a stateless bijection preserves them. A translator that detects an unavailable phase is not stateless and must be charged as a simulator. The proposition does not characterize the full minimal automaton in general. □

Proof of Theorem 3.1. Lemma 4.2 and minimization over the fixed dictionary give the checkpoint lower. At least one state is live, so an M-state autonomous observer has at most $M-1$ terminal values. Use Proposition 4.4 for the sharpened width when its witness is θ_0. Lemmas 4.1 and 4.3 give the common upper and attainment. Its checkpoint version simply uses the same program with the exported dictionary. If $D_0 = 1$, its width agrees with the lower and proves (3.5). Under the witness, one productive call is necessary and costs at least κ in expectation, so budgets below κ are infeasible. The mandatory live states and at least one terminal state prove the other infeasibility assertion. □

5. Raw balance and stability without a variance floor

A finite moment criterion. Let E be a finite-dimensional real ordered space with generating positive cone and a strictly positive mass functional u. Its normalized positive base $S = \{s \ge 0 : u(s) = 1\}$ is compact. This is a verification device for the following proposition; Theorem 3.1 does not assume a finite-dimensional physical space.

For each of finitely many allowed within-call commands j, suppose the raw joint report–audit law depends linearly and positively on $s \in S$. After the declared report transformation, and a task isometry $T_j : H \to H$ with $T_jH = H$, write $\widetilde{Z} = T_j^{-1}Z$. The following vector measures are linear functionals of s:

$$A_j(D)(s) = \mathbb{E}_s[\mathbf{1}_{\{x\in D\}}], \quad B_j(D)(s) = \mathbb{E}_s[\mathbf{1}_{\{x\in D\}}\widetilde{Z}], \quad C_j(D)(s) = \mathbb{E}_s[\mathbf{1}_{\{x\in D\}}\|\widetilde{Z}\|^2].$$

Here B_j has k coordinates. Fix a genuinely executed reference law K_0 and its moments (2.1).

Proposition 5.1 (Raw task-moment balance criterion)**.** *A fixed distribution $\lambda \in \Delta(J)$ balances report and squared-loss measures for every normalized input s if and only if, as E^*-valued measures,*

$$\sum_j \lambda_j A_j(dx) = \nu(dx)u, \tag{5.1}$$

$$\sum_j \lambda_j B_j(dx) = f(x)\nu(dx)u, \tag{5.2}$$

$$\sum_j \lambda_j C_j(dx) = (w(x) + \|f(x)\|^2)\nu(dx)u. \tag{5.3}$$

Equality on a countable determining algebra of report events suffices. The set of such λ is compact and convex. If it is empty, finitely many scalar moment constraints already separate all mixtures in $\Delta(J)$ from the required identities.

Proof. Expand $\|\widetilde{Z} - r\|^2 = \|\widetilde{Z}\|^2 - 2r \cdot \widetilde{Z} + \|r\|^2$. The three identities give (2.3)–(2.4), using that T_j is an isometry. Conversely, the mass identity and equality of this quadratic polynomial for r in the affine hull of H determine the first moments in that affine hull and the squared-norm moment. One may first translate the affine hull to the origin and use an orthonormal coordinate system on it; orthogonal coordinates of Z are then fixed and carry no additional constraint. Equality for all $s \in S$ is equality as linear functionals because the positive cone is generating. Uniqueness of finite signed measures extends equality from a determining algebra to all Borel sets. Each coordinate constraint is affine and closed in the finite simplex. Compactness and the finite-intersection property give a finite inconsistent subfamily if no mixture exists. Its affine image of the simplex is compact and convex and omits the prescribed moment vector. Finite-dimensional strict separation proves the last assertion. This is a certificate of impossibility for the declared command family, not for arbitrary causal algorithms. □

The criterion is insensitive to density zeros or changes in the rank of a posterior. It also shows why report-only averaging is insufficient. A report identically equal to zero has the same law for hidden states $e = \pm 1$, but an audit $Z = e$ has loss $(e - r)^2$. Without balancing its first moment there is no common d_r valid for all outputs, despite perfect report balance.

A finite physical randomization satisfying $\sum_j \lambda_j G_j(s) = \omega u(s)$ is one sufficient source of (5.1)–(5.3) for any subsequent fixed instrument. More generally, it suffices to annihilate input dependence only on the report, first-moment and squared-norm functionals; the entire hidden state need not be erased. These are linear moment identities, not an assertion that a Fisher matrix, covariance or fixed-rank parametrization is primitive. Randomization itself is standard [3, 4]; the stopped conclusion follows only after the actual witness, paid acquisition and exported readout conditions have also been verified.

Integrated predictive-update stability. The useful continuity is that of unnormalized instruments. Uniform pointwise continuity of a normalized posterior can fail at zero-mass reports.

Proposition 5.2 (Integrated stability through singular reports)**.** *Let a classical stochastic instrument, or a finite-dimensional completely positive trace-preserving instrument, map two normalized inputs s, t to subnormalized post-report states A_x, B_x with respect to a common finite dominating measure. Put $p_x = u(A_x)$, $q_x = u(B_x)$, and normalize on positive-mass reports. In the classical ℓ^1*

norm or quantum trace norm,

$$\int \|A_x - B_x\|_1 \, dx \le \|s - t\|_1, \tag{5.4}$$

$$\|p - q\|_{\mathrm{TV}} \le \tfrac{1}{2}\|s - t\|_1, \tag{5.5}$$

$$\int \min(p_x, q_x) \left\| \frac{A_x}{p_x} - \frac{B_x}{q_x} \right\|_1 dx \le 2\|s - t\|_1. \tag{5.6}$$

Zero-mass terms are defined to contribute zero. The same inequalities apply to a complete finite causal continuation instrument under the same fixed parameter-independent policy. If its stopped scored instrument terminates almost surely under both inputs, the unnormalized contraction and report bound extend to that stopped instrument.

Proof. In the quantum case write the Hermitian difference $s - t = D_+ - D_-$ with positive parts. Positivity, the triangle inequality and preservation of integrated trace give $\int \|\mathcal{I}_x(D_+ - D_-)\|_1 dx \le \operatorname{tr} D_+ + \operatorname{tr} D_- = \|s - t\|_1$. The same Jordan-decomposition argument works for classical signed measures. Taking masses proves (5.5). When $p \le q$,

$$p\|A/p - B/q\|_1 = \|A - (p/q)B\|_1 \le \|A - B\|_1 + (q - p).$$

Interchange A, B for the other case and integrate, using $|p - q| \le \|A - B\|_1$. Composition of legal physical instruments remains positive and trace preserving, so the finite-continuation assertion follows. For stopped paths, sum over commitment time, first to a finite horizon while retaining the surviving output. This is a positive mass-preserving instrument. Almost-sure termination under both inputs lets the surviving masses tend to zero; monotone summation gives the stopped assertion. No bound on expected stopping time follows from this argument; it requires a separate cost certificate. □

This stability controls the *actual report mass* together with the posterior, including pure states and likelihood zeros. It is not a contraction factor strictly below one, and no normalized posterior is stored by the finite-state upper. In particular a pointwise variance-floor constant diverging along nearly pure histories does not force the risk constant of Theorem 3.1 to diverge.

6. Actual marked geometry, singular supports and atoms

The following calculation is inherited from the stopped-experiment and predictable-geometry results [1, 2]. It is included to make the sharp consequences independent of those manuscripts.

Let $\sigma = (f, \sqrt{w})_{\#}\nu$, supported on $K \times [0, 1]$ with compact $K \subset H$. Suppose there are $d, \alpha > 0$, $0 < r_0 \le 1$ and constants $c, C > 0$ such that for every $x \in K$, $0 < r \le r_0$ and $0 < s \le 1$,

$$c r^d s^\alpha \le \sigma(B(x, r) \times [0, s]) \le C r^d s^\alpha. \tag{6.1}$$

These are statements about the *actual* report–task pushforward, not the volume of a reachable parameter set. They include Ahlfors-regular singular location measures of noninteger d and a reliability coordinate with exponent α.

Proposition 6.1 (Marked acquired profile)**.** *Under* (6.1)*, for all $b \ge 1$ and $m \ge 1$,*

$$\mathcal{Q}_m(b) \asymp \max\{b^{-2/\alpha}, (bm)^{-2/(d+\alpha)}\}, \tag{6.2}$$

with constants depending only on the displayed geometric data and the bounded location set.

Proof. A finite cover of K by balls of radius $r_0/2$ gives $\sigma(K \times [0, s]) \le C_0 s^\alpha$. Under any density cap b, the selected reliability obeys $\mathbb{P}(\sqrt{w} \le s) \le b C_0 s^\alpha$. Choose $s = (2C_0 b)^{-1/\alpha}$, decreasing it by a fixed factor if necessary, to obtain the first lower bound. For any dictionary of m points, the sublevel $w + \operatorname{dist}(f, C)^2 \le t^2$ is contained in m rectangles of location radius t and reliability at most t. Centers outside K can be moved for this covering argument to a point of K within t, increasing the radius to $2t$; empty balls are discarded. For $2t \le r_0$ its mass is at most $C_1 m t^{d+\alpha}$. Taking

$t = c_1(bm)^{-1/(d+\alpha)}$ with a sufficiently small fixed c_1 bounds the selected sublevel mass by $1/2$ and gives the second lower. This choice works for all $b, m \geq 1$ after reducing c_1 once.

For the upper, the marginal $\sigma(\cdot \times [0,1])$ satisfies two-sided d-dimensional ball bounds. Maximal separated sets therefore provide at least $c_2 r^{-d}$ centers at separation $4r$ for small r. Choose

$$j = \min\{m, \lfloor c_3 b^{d/\alpha} \rfloor\}, \qquad t = C_2(bj)^{-1/(d+\alpha)}.$$

Take $c_3 > 0$ sufficiently small and C_2 sufficiently large, in that order with the packing inequality also enforced by decreasing c_3. For $b \geq b_0$, where

$$b_0 = \max\{1, (2/c_3)^{\alpha/d}, (C_2/\min(r_0/8, 1))^{d+\alpha}\},$$

we have $j \geq 1$, $t \leq \min(r_0/8, 1)$, and j disjoint rectangles $B(x_i, t) \times [0, t]$ whose union has mass at least $cjt^{d+\alpha} \geq 1/b$. Here the required packing condition $jt^d \leq c_4$ follows from $j \leq c_3 b^{d/\alpha}$. A density of mass one bounded by b can be supported on this union (constant inverse union mass suffices), and its distortion to these centers is at most $2t^2$. Substituting j gives the upper in (6.2). For $1 \leq b < b_0$, accept every report and use any fixed point in H. This gives a uniform upper while $b^{-2/\alpha} \geq b_0^{-2/\alpha}$ supplies a uniform positive comparison, independently of unbounded m. No compactness assertion about the integer m is needed. □

Proposition 6.2 (An exact acquired atom transition)**.** *Suppose the marked law is $\sigma = \lambda\delta_{(x_*,0)} + (1-\lambda)\sigma_0$, $0 < \lambda < 1$, where σ_0 is a probability law and has no atom at $(x_*, 0)$. Write $\mathcal{Q}^0_m$ for its profile. If $b\lambda \geq 1$, $\mathcal{Q}_m(b) = 0$ for every $m \geq 1$. If $r = 1 - b\lambda > 0$, then for every $m \geq 2$,*

$$r\mathcal{Q}^0_m\left(\frac{b(1-\lambda)}{r}\right) \leq \mathcal{Q}_m(b) \leq r\mathcal{Q}^0_{m-1}\left(\frac{b(1-\lambda)}{r}\right). \tag{6.3}$$

Consequently the exact case of Theorem 3.1 retains the zero-risk threshold $N = \kappa/\lambda$, with no unknown mass renormalization.

Proof. Above threshold, select only the actual atom and output x_*. Below threshold, any selected probability has remainder mass $s \geq r$. With its fixed dictionary, scale its remainder submeasure by r/s. This yields a submeasure of mass r, still dominated by $b(1-\lambda)\sigma_0$, and with no larger nonnegative distortion than the original remainder part. Its minimum possible distortion over dictionaries of size m is the left side of (6.3). For the upper, select the full allowable atom mass $b\lambda$, allocate one center to x_*, and use an optimal $(m-1)$-point dictionary on a mass-r submeasure of the remainder. The unnormalized factor r is retained throughout. □

The atom result applies to any remainder, including a mixture of singular strata. It does not assert that arbitrary changing-rank geometries satisfy (6.1). In the raw quantum construction below, rank changes can occur while the task law still satisfies these rectangle estimates; this is proved directly rather than inferred from ambient matrix dimension.

7. Two raw realizations without a positivity floor

Paid, history-dependent heralding. There are two physically distinct ports. At port G a gate call costs one. Its success probability, conditional on every hidden and observed past, belongs to $[p_0, 1]$, where $p_0 > 0$ is a known guaranteed lower bound, not the unknown true rate. Success permits one call at acquisition port A, costing one. No commitment is legal at G. After rejecting at A, the protocol returns to G. No hidden state is reset by this routing convention. The class includes a reference model with constant gate success p_0 and the productive laws specified below.

Lemma 7.1 (The exact robust paid-attempt scale)**.** *For this interface set $\kappa = 1 + 1/p_0$. Every all-history policy under the reference model satisfies $\mathbb{E}C = \kappa\mathbb{E}A_\tau$. The program which calls the legal current port reaches one productive report at conditional expected cost at most κ, for every model and history. Both ports require distinct live states, so $B = 2$ is a semantic mandatory lower. Robust expected-budget feasibility is exactly $N \geq \kappa$ when at least three observer states are available.*

Proof. Let G_τ, A_τ be gate and productive counts. Every productive call consumes one gate success, and a terminating path has no unused success. Predictable counting gives $\mathbb{E}A_\tau = p_0\mathbb{E}G_\tau$ under the witness, without requiring independent cycles. Hence $\mathbb{E}C = (1 + 1/p_0)\mathbb{E}A_\tau$. For the upper, the probability of j consecutive failures, conditional on any history at the start, is at most $(1 - p_0)^j$. Summing gives at most $1/p_0$ gate calls and one acquisition call. The physical gate and acquisition ports have disjoint legal command sets, and both occur before any legal commitment under the same witness. Proposition 4.4 applies. Accepting the first productive report gives the feasibility converse. At $p_0 = 1$ the cost is two, not one. At fixed N, the task is infeasible when $1 + 1/p_0 > N$; rare-rate uniformity always refers to the feasible rescaled budget N/κ. □

Controlled classical hidden states. Let the physical state contain a sign $E \in \{-1, 1\}$ and an arbitrary nuisance state. Between productive calls any causal stochastic or deterministic evolution is allowed, with no mixing or positivity requirement. The observer may choose either of two physical commands on the sign: identity or sign flip, followed immediately by the same known observation-and-audit instrument. There is no unknown intervening evolution between this chosen command and that instrument. The physical task is to predict the *post-command* audit, not a pre-intervention sign.

For input sign e, let $J_e(dx, dz)$ be any fixed probability kernel with bounded audit in H; it can have atoms, singular parts and sign-dependent report likelihoods. Nuisance evolution is unrestricted provided this productive instrument depends on the sign through these stated kernels. The class contains a reference member which supplies a fresh fair sign before each productive command, independently of all prior reports, and uses the constant gate rate p_0. This is an admitted physical preparation model, not a reset available freely to every observer. Its productive law is

$$K_0 = \tfrac{1}{2}(J_+ + J_-). \tag{7.1}$$

Theorem 7.2 (Classical task balance at zero positivity floor)**.** *The classical family satisfies Theorem 3.1 with $B = 2$, $D_0 = 1$ and $\kappa = 1 + 1/p_0$, for the task geometry derived from* (7.1)*. In particular, for all $N \geq \kappa$ and $M \geq 3$,*

$$R^{\Theta}_{\mathrm{aut}}(N, M) = \mathcal{Q}_{M-2}(N/\kappa), \qquad R^{\Theta}_{\mathrm{cp}}(N, m) = \mathcal{Q}_m(N/\kappa). \tag{7.2}$$

The constants require no positive lower bound on transition probabilities or conditional residual variances.

Proof. The upper chooses identity or flip with a fresh fair coin within the productive command. For every conditional pre-command sign distribution, averaging makes the post-command sign fair. By linearity the joint report–audit law, not merely its report marginal, is exactly K_0. Take the identity export map. The nuisance state need not be reset or stored. Under the reference member, each chosen physical command, even conditioned on its chosen coin and on the full preceding history, still has law K_0; hence arbitrary competitors satisfy the reference conditions. Lemma 7.1 gives both cost directions and the mandatory states. Theorem 3.1 proves (7.2). Classical instrument continuity follows from Proposition 5.2, including deterministic transitions. □

For a concrete singular geometry, draw (Y, T) independently of the sign with any probability ν_0 on $K \times [0, 1]$, and let $Z = Y + TEe_1$ in a fixed compact convex carrier. The reference has $f = Y, w = T^2$. Choose a d-regular singular measure on K and reliability law $\mathbb{P}(T \leq s) = s^\alpha$ to obtain (6.1); add a genuinely prepared atom $(Y, T) = (x_*, 0)$ to obtain Proposition 6.2. The report can instead be informative: a weak sign report V with $\mathbb{P}(V = v \mid E = e) = (1 + \eta ve)/2$ gives $f = Y + \eta VTe_1$ and $w = (1 - \eta^2)T^2$ under the reference. Its profile is still exact, although rectangle estimates must then be checked for that sheared marked law rather than assumed. In contrast, a static known-to-the-history sign can have zero conditional variance at individual histories. Such histories violate a positive pointwise variance mark but do not invalidate this minimax theorem.

Noncommuting quantum instruments with rank changes. Consider a qubit with arbitrary unknown completely positive trace-preserving evolution between port interactions. No depolarizing component, spectral gap or full-rank floor is imposed. Gate instruments have success effects bounded below by $p_0 I$; their surviving states may be pure. The productive port permits a Pauli command from $\{I, \sigma_x, \sigma_y, \sigma_z\}$, followed by a fixed known instrument and audit, with no unknown operation inserted inside this command. The class contains a reference member that prepares $I/2$ immediately before every productive Pauli command and has gate success p_0.

Here is an explicit noncommuting instrument. Draw (y, t) with any fixed probability ν_0 on $K \times [0, 1]$. Choose a Borel angle $\phi(y, t)$ and $0 \le \eta(y, t) \le 1$. Set

$$n = (\cos\phi, \sin\phi, 0), \qquad E_v = \tfrac{1}{2}(I + v\eta n \cdot \sigma), \quad L_v = \sqrt{E_v}, \qquad v = \pm 1.$$

The unnormalized post-report state is $L_v \rho L_v$, and the report is $x = (y, t, v)$. At commitment measure σ_z, with result $e = \pm 1$, and score $Z = y + tee_1$. The axes may vary, so effects at different reports need not commute.

Theorem 7.3 (Quantum verification through pure posterior strata)**.** *For the stated quantum family, the same conclusions* (7.2) *hold with the actual reference mark law* $\nu = \nu_0 \otimes \frac{1}{2}(\delta_{-1} + \delta_1)$ *and* $f = y, w = t^2$*. They remain valid at* $\eta = 1$*, at pure incoming states, and across report strata where the posterior rank changes. If* ν_0 *has the geometry or atom described above, the corresponding regular or exact atomic risk law follows from the* same *central theorem.*

Proof. Writing $\rho = (I + r \cdot \sigma)/2$ shows directly that

$$\tfrac{1}{4} \sum_{U \in \{I, \sigma_x, \sigma_y, \sigma_z\}} U \rho U^* = I/2.$$

This standard Pauli averaging identity [4] is a charged within-call preparation, not free additional quantum memory. Every unnormalized report–audit probability is linear in ρ, so its average is its value at $I/2$. The reference member is invariant under *each* chosen Pauli command, including after conditioning on that command and all earlier observations. Thus it gives the required reference for all competitors, while the random command gives one common upper for every unknown input state.

Under $I/2$, the probability of v is $\mathrm{tr}(E_v)/2 = 1/2$ and the normalized post-report state is E_v. Its z component is zero, so the subsequent Pauli audit has both signs with probability $1/2$. This gives the stated f, w and the full unnormalized loss identity. The eigenvalues of E_v are $(1 \pm \eta)/2$. Choosing $\eta = 1$ on one set of positive ν_0-mass and $\eta = 1/2$ on another gives actual rank-one and rank-two strata, not merely a formal rank-deficient limit. Different equatorial axes give noncommuting effects. Proposition 5.2 supplies report continuity and actual-mass-weighted update stability without dividing by a positive lower eigenvalue. Lemma 7.1 and Theorem 3.1 complete the proof. □

The quantum carrier here is finite dimensional and the observer is classical. It has two live labels and its terminal dictionary, not an uncharged quantum register. Interventions change the declared post-command audit; neither theorem asserts that randomizing an unknown pre-intervention quantity preserves its original estimation task. Known classical or quantum preparation operations, their costs and the availability of the reference member are explicit hypotheses, not automatic consequences of positivity.

8. A nonintervening Gaussian sensing law

In this section the task is not randomized by a physical intervention. At a productive call a current hidden sign $E \in \{-1, 1\}$ emits

$$X = aE + \sigma G, \qquad a, \sigma > 0, \tag{8.1}$$

where G is a fresh standard normal variable. The audit is that current sign E. All history-dependent hidden sign evolutions are allowed, including the static signs and an independently refreshed fair sign. No command may alter the sign between the sensor call and its audit. The heralding interface is that of Lemma 7.1; all three indicated sign models with constant gate rate p_0 are admitted. Thus arbitrary old observations or gate histories may be informative, but the class has no positive uniform conditional variance floor.

Under the independent fair reference, the actual productive density and conditional moments are

$$\nu(dx) = \frac{\varphi_\sigma(x-a) + \varphi_\sigma(x+a)}{2}dx, \quad f(x) = \tanh(kx), \quad w(x) = \operatorname{sech}^2(kx), \quad k = a/\sigma^2. \tag{8.2}$$

The formulas follow by dividing the two Gaussian likelihoods; their ratio is $\exp(2kx)$. For $b \geq 1$, let $t_b \geq 0$ be the unique threshold with $\nu(|X| \geq t_b) = 1/b$, taking $t_1 = 0$, and define

$$D(b) = b\int_{|x|\geq t_b} w(x)\nu(dx), \qquad U(b) = 2b\int_{|x|\geq t_b}(1-|f(x)|)\nu(dx). \tag{8.3}$$

The two-sided threshold construction and the logarithmic waiting scale were already proved for a positive-prior-floor class in [1]. Here the lower is instead supplied by one admitted fair reference member: this removes that floor, permits static hidden signs, and yields a ratio-one minimax asymptotic uniformly over every larger terminal alphabet. The conditional-variance selection principle below is standard in regression with rejection [6], with the earlier recognition tradeoff in [5]. The assertion here additionally constrains the paid stopped experiment and all of its persistent states; it does not assert priority for a threshold rule.

Theorem 8.1 (Terminal-value transition and the sharp waiting asymptotic)**.** *Put $b = N/\kappa$, with $\kappa = 1 + 1/p_0$, and assume $N \geq \kappa$. In the autonomous model,*

$$R^{\Theta}_{\mathrm{aut}}(N,3) = 1, \qquad D(b) \leq R^{\Theta}_{\mathrm{aut}}(N,M) \leq U(b) \leq 2D(b) \quad (M \geq 4). \tag{8.4}$$

Fewer than three states are infeasible. For checkpoint observers, the same statements hold with one versus at least two terminal values. Uniformly over all integers $M \geq 4$,

$$R^{\Theta}_{\mathrm{aut}}(N,M) \sim 4\exp(-2a^2/\sigma^2)\exp\left[-\frac{2a}{\sigma}\sqrt{2\log(N/\kappa)}\right] \quad \text{as } N/\kappa \longrightarrow \infty. \tag{8.5}$$

The same uniform asymptotic holds over checkpoint dictionaries of size at least two. It is not a fixed-budget limit as $p_0 \downarrow 0$.

Proof. The lower uses the independently refreshed fair sign, not a chosen posterior prior for each competitor. Lemma 4.2 bounds the terminal X-law by $b\nu$, and the conditional residual variance in (8.2) is unavoidable even with infinitely many output values. Since w decreases with $|x|$, its minimum integral against a mass-one density bounded by b is $D(b)$. For completeness, subtract the threshold rule from any feasible density: on $|x| \geq t_b$, $w(x) \leq w(t_b)$ and $h - b \leq 0$; on the complement, $w(x) \geq w(t_b)$ and $h \geq 0$. Integrating the product and using zero total mass difference proves this minimization without a restriction to threshold competitors.

For the upper, accept exactly when $|X| \geq t_b$ and output $\operatorname{sign}(X)$. Reflection of the Gaussian kernel shows that its acceptance probability is $1/b$ for each fixed current sign, and its unnormalized squared-loss contribution is $U(b)/b$ for each sign. The same identities hold for every mixture of signs conditional on any earlier history. The proof of Lemma 4.3 therefore gives expected cost at most κb and risk $U(b)$, without refreshing or estimating the hidden sign. This uses two live port labels and the two outputs ± 1. Algebra gives

$$\frac{2(1-|f|)}{1-f^2} = \frac{2}{1+|f|},$$

so $D(b) \leq U(b) \leq 2D(b)$ and, when $t_b > 0$, $U(b)/D(b) \leq 2/(1+\tanh(kt_b))$.

If there is only one terminal value $r \in [-1,1]$, the two admitted static-sign models give worst-case risk at least $\max\{(1-r)^2,(-1-r)^2\} \geq 1$, irrespective of stopping. Outputting zero at the first productive report attains one. The mandatory physical ports leave only one terminal value when $M = 3$. This argument allows all fresh randomized stopping and routing, not only thresholds.

It remains to identify the asymptotic constant. Let $u_b = (t_b - a)/\sigma$ and $c = a/\sigma > 0$. The tail equation is

$$b^{-1} = \overline{\Phi}(u_b) + \overline{\Phi}(u_b + 2c).$$

Integration by parts gives, for $u > 0$, $\frac{u}{1+u^2}\varphi(u) \leq \overline{\Phi}(u) \leq \varphi(u)/u$. The second tail is negligible relative to the first, so $u_b = \sqrt{2\log b} + O(\log\log b/\sqrt{\log b})$. In particular $t_b = a + \sigma\sqrt{2\log b} + o(1)$. For any fixed $v > 0$, the same Gaussian tail bounds show $\nu(|X| > t_b + v)/\nu(|X| \geq t_b) \to 0$. Thus $|X| - t_b \to 0$ in probability under the selected tail law. Moreover $0 \leq w(X)/w(t_b) \leq 1$ there, and this ratio tends to one in that probability. Bounded convergence in probability yields $D(b)/w(t_b) \to 1$. Since $w(t_b) \sim 4e^{-2kt_b}$, the right side of (8.5) follows. Finally $U(b)/D(b) \to 1$. The lower and upper do not depend on $M \geq 4$, proving the claimed uniformity. □

This application is also a raw transformed-task verification of Theorem 3.1. After a sensor report, choose a fresh fair $g \in \{-1,1\}$, use virtual pair (gX, gE) and export $T_g r = gr$. Averaging yields exactly the reference joint law for every pre-sensor sign distribution, with $D_0 = 2$. No physical randomization of the target occurs. The symmetric threshold rule above removes even the virtual coin. Its sharp asymptotic arises from the actually acquired Gaussian tails, not a formal one-dimensional parameter space. The one-value obstruction requires the two static models in addition to the fair lower witness; it is not inferred from a variance-floor hypothesis.

9. Finite descriptions and stopped numerical certificates

An arbitrary Borel acceptance set is not a finite computer program. We first state the error bound on the scale that a stopped experiment actually uses.

Proposition 9.1 (Relative-acceptance implementation)**.** *Consider an implemented attempt program with the same legal physical ports and B live states as the ideal one. On every surviving pre-attempt history suppose its conditional expected cost is at most $\widetilde{\kappa}$, its acceptance probability is at least $(1-\rho)/b$, and its unnormalized terminal loss is at most $(Q+\eta)/b$, where $0 \leq \rho < 1$, $\eta \geq 0$, $b \geq 1$. It then terminates almost surely and satisfies*

$$\mathbb{E}C \leq \frac{\widetilde{\kappa} b}{1-\rho}, \qquad R \leq \frac{Q+\eta}{1-\rho}. \tag{9.1}$$

The terminal alphabet is the actual exported dictionary. These bounds remain valid for unknown history-dependent implementation errors satisfying the stated conditional certificates.

Proof. The probability of surviving j attempts is at most $(1-(1-\rho)/b)^j$. Predictable summation of the bounded conditional costs and nonnegative terminal-loss contributions proves both inequalities and finite expected attempts. Each attempt terminates almost surely by its own cost certificate. No comparison of terminal distributions was used to infer duration. □

For example, an absolute one-attempt acceptance error e and unnormalized loss error e_L give $\rho = be$ and $\eta = be_L$, not $\rho = e$ and $\eta = e_L$. Rare selection amplifies these absolute errors. A task-readout perturbation of size δ in a carrier of diameter D_H changes loss by at most $2D_H\delta + \delta^2$ *conditional on acceptance*; its unnormalized error is weighted by acceptance mass. A continuous report and a digitized atomic report need not be close in total variation; boundary-event or unnormalized moment estimates must be supplied instead.

Proposition 9.2 (Finite-description upper at the same cost)**.** *In addition to exact balance, assume the known reference measure admits an effective sequence of finite physical report partitions with*

certified approximation of bounded acceptance functions in $L^1(\nu)$. Assume finite points in H and their exported values can be approximated within H with certified uniform error, and their primitive report classifiers and probabilities have declared finite descriptions, execution costs and integration error bounds. Then, for every $b \geq 1$, $m \geq 1$ and $e > 0$, there is a finite-description observer with at most $B + D_0 m$ states, cost at most κb and risk at most $\mathcal{Q}_m(b) + e$. This is an existence and approximation result under effective certificates, not a polynomial-time discovery theorem for arbitrary kernels.

Proof. For $b > 1$, choose $1 < b' < b$. By Lemma 4.1, $\mathcal{Q}_m(b')$ can be made arbitrarily close to $\mathcal{Q}_m(b)$. Approximate its optimal finite dictionary inside H and use its nearest-center cells. Approximate the bounded acceptance h/b' and this finite center assignment jointly on a finite partition. The assumptions supply certified $L^1(\nu)$ error bounds. Because all squared losses are bounded and dictionary errors change them uniformly, the acceptance and unnormalized loss errors tend to zero. The ideal acceptance $1/b'$ has a strictly positive margin above $1/b$. Take the approximation sufficiently accurate that its acceptance is still at least $1/b$. Balance makes these quantities the same on every surviving history and model. Its risk is its unnormalized loss divided by acceptance; it tends to the ideal risk. The attempt count proof then gives cost at most κb. For $b = 1$, accept the first report identically and approximate only the dictionary and assignment; no acceptance margin is needed. Current partition indices and command coins are scratch variables, and only the routing or exported output label persists.

The argument is constructive once the indicated approximation and integration certificates are supplied. A finite search over candidate dictionaries and report partitions, with bounded integration intervals, can certify an available upper. No claim that a local optimizer supplies a global lower is involved. Fully effective global convergence additionally uses effective compactness of the dictionary carrier and uniform integration bounds; a finite dictionary net and the Lipschitz loss then give finite lower and upper profile approximations, potentially at exponential search cost. □

For regular geometry, finitely supplied centers and coordinate certificates give a direct upper without learning an arbitrary acceptance density. Replace inner rectangles from Proposition 6.1 by outward rounded rectangles, accepting every true inner rectangle and only points with marked distance at most a fixed multiple of its radius. Actual mass remains at least $1/b$ and distortion remains of the same order. With m centers in a fixed k-dimensional box and radius t, $p = O(1 + \log(1/t))$ precision bits suffice when the coordinate evaluation has that error certificate. A sequential implementation uses $O(mkp)$ read-only data bits, $O(kp+\log m)$ scratch bits and $O(mkp^2)$ elementary arithmetic operations per productive report, in addition to the specified physical evaluation and command costs. These are sufficient costs, not optimal bit/time lower bounds.

An end-to-end dyadic example at zero positivity floor. Use the classical family of Theorem 7.2 with independent uniform $(Y, T) \in [0, 1]^2$, post-command audit $Z = Y + TE$, and $p_0 = 1/4$. Thus $\kappa = 5$. Use three states: gate, acquisition, and terminal value $1/2$. Ideally accept

$$A = [3/8, 5/8] \times [0, 1/8].$$

Under balanced task measures,

$$\nu(A) = 1/32, \qquad \int_A [(y - 1/2)^2 + t^2]\, dy\, dt = 1/3072. \tag{9.2}$$

Let $e = 2^{-14}$, and suppose the physical digitizer supplies clipped coordinates with errors at most e, using only (Y, T) and not the discarded sign-flip coin. Accept when

$$3/8 + e \leq \widehat{Y} \leq 5/8 - e, \qquad \widehat{T} \leq 1/8 - e.$$

The accepted true report lies in A, and every point in $A_{\mathrm{in}} = [3/8 + 2e, 5/8 - 2e] \times [0, 1/8 - 2e]$ is accepted. Put

$$\alpha_- = (1/4 - 4e)(1/8 - 2e), \quad \beta_+ = 1/3072, \quad W_G = 5/\alpha_-, \quad W_A = W_G - 4, \quad U_G = U_A = \beta_+/\alpha_-. \tag{9.3}$$

These are exact rational numbers. The reference rows with gate success $1/4$ and productive acceptance α_- obey $W \geq \mathbf{1} + QW$ and $U \geq \beta + QU$. Every actual gate row has greater success and every implemented productive row has acceptance at least α_- and loss contribution at most β_+. Thus the same inequalities hold directly for the implemented raw rows: a killed-potential certificate has relative perturbation constants $\rho = \eta = \zeta = 0$ relative to these conservative rows. Summing the finite-horizon inequalities, *including* the nonnegative surviving W and U terms before taking the limit, proves

$$\mathbb{E}C \leq 5/\alpha_- < 161, \qquad R \leq 1/(3072\alpha_-). \tag{9.4}$$

Equivalently these bounds follow from Proposition 9.1 with the conservative acceptance and numerator. A same-report changed decision lies only in $A \setminus A_{\mathrm{in}}$, whose probability is at most $1/32 - \alpha_-$. No total-variation comparison of continuous and atomic report laws was made.

For a matching all-program lower at the same budget 161 and three states, Theorem 3.1 gives $\mathcal{Q}_1(161/5)$. For any output r, a sublevel $t^2 + (y - r)^2 \leq z$ in $[0, 1]^2$ has area at most $\pi z/2$. Integration of the selected tail gives $\mathcal{Q}_1(b) \geq 1/(\pi b)$. When $b \geq 8/\pi$, a half-disk centered at $(1/2, 0)$ of area $1/b$ stays inside the square and attains equality. Consequently

$$\frac{5}{161\pi} = R^{\Theta}_{\mathrm{aut}}(161, 3) \leq R_{\mathrm{dyadic}} \leq \frac{1}{3072\alpha_-}. \tag{9.5}$$

The equality is the ideal Borel optimum; the displayed dyadic observer is a separately certified nearby finite implementation.

The three threshold integers are $6145, 10239, 2047$, requiring 45 read-only bits at 15 bits each, plus the fixed comparator program and output $1/2$. Two 15-bit digitized-coordinate registers, one fresh flip bit and fixed control scratch suffice within a productive call. The three persistent states require two encoding bits; neither a counter nor a retained random seed is present. Comparisons use $O(14)$ bit operations per productive call. If a gate call costs at most t_G time and productive preparation, digitization and comparison together cost at most $t_A(14)$, expected physical execution time is at most $161 \max(t_G, t_A(14))$. The promised digitizer and balanced physical pulse costs are not free. One subsequent physical audit makes the total expected raw-call budget at most 162.

10. Causal simulation and a joint attained-error law

Proposition 10.1 (Typed common-task composition)**.** *A simulator from a source experiment to a target experiment must specify a parameter-independent strategy lift, report transformation, internal state, clock, task readout and physical-command compatibility. Suppose it uses at most S retained states, a target observer uses M, and their entire stopped scored transcript laws differ by at most ε in total variation, with convention $\|P - Q\|_{\mathrm{TV}} = \sup_A |P(A) - Q(A)|$. For loss in $[0, L]$, the composed observer uses at most SM states and its risk differs by at most $L\varepsilon$. Its expected cost bound requires its own additive-cost or killed-potential certificate. Two compatible causal simulators have total deficiency at most the sum of their deficiencies and a product state upper bound.*

Proof. The simulator and observer retain their ordered pair of states; clocks and any continuing random seeds are part of those states. Compatibility makes the composition a legal causal procedure. For a common bounded scored loss, the layer-cake formula and the total-variation convention give the risk bound. Contractivity of total variation under a common Markov kernel and the triangle inequality give the two-simulator bound on complete stopped scored laws. Nonnegative costs must instead be summed along actual executed calls or bounded by separately certified stopped potentials.

A small terminal-law discrepancy alone can coexist with arbitrarily large expected waiting. Reverse risk bounds need a reverse simulator or a separate attained lower construction. □

The state product is an implementation upper, not a claim of optimal simulator size. Root-mean-square readout errors add by the triangle inequality before squaring; deterministic errors may alternatively use the bounded-loss estimate following Proposition 9.1. Neither convention makes an allowed upper bound on calibration or simulation error an unavoidable risk lower.

Theorem 10.2 (Same-task attained calibration and erasure)**.** *Use the exact balanced class $D_0 = 1$, $B = 2$. Add an initial paid observation of an independent fair sign E_0 erased with probability 2ε, where $0 \le \varepsilon \le 1/2$, and a final paid audit of*

$$(Z, E_0, s\delta), \qquad s \in \{-1, 1\}, \quad \delta \ge 0.$$

The fixed unknown sign s affects no pre-audit kernel. Conditional on the initial reported tag, the entire core experiment contains no additional information about E_0 on the erased branch; in particular its physical kernels may not read that hidden sign. All output coordinates are committed before this audit. Use squared Euclidean loss on the product carrier. For $N \ge \kappa + 2$ and $M \ge 13$, put

$$b = (N-2)/\kappa, \qquad m = \lfloor (M-1)/3 \rfloor - 2.$$

Then

$$\mathcal{Q}_M(b) + 2\varepsilon + \delta^2 \le R^{\Theta}_{\mathrm{joint}}(N, M) \le \mathcal{Q}_m(b) + 2\varepsilon + \delta^2. \tag{10.1}$$

If (6.1) *holds, this is a matching law*

$$R^{\Theta}_{\mathrm{joint}}(N, M) \asymp F((N-2)/\kappa, M) + \delta^2 + \varepsilon, \tag{10.2}$$

with the geometry constants specified earlier. The early report channel has causal deficiency exactly *ε relative to the perfect-sign channel when its virtual sign must be delivered before any core call.*

Proof. For the upper, use one initial state and three tagged copies of the $2 + m$-state core observer: tags $+$, $-$ and erasure. Their number is $1 + 3(2 + m) \le M$. Output the reported sign on known branches, zero on erasure, and zero for the inaccessible calibration sign. The balanced core upper has the same risk after every preceding history, hence its contribution is $\mathcal{Q}_m(b)$. The other contributions are exactly 2ε and δ^2. Initial observation and final audit each cost one.

For the lower, use the fixed reference core. Expected core cost is at most $N - 2$, and its terminal projected dictionary has at most M values, so Lemma 4.2 gives $\mathcal{Q}_M(b)$. On the erased branch, induction over the causal core kernels and independent observer coins leaves E_0 fair conditional on all pre-audit information. Any sign estimate therefore has conditional squared loss at least one. Finally average the two indistinguishable alternatives $s = \pm 1$: for any committed last coordinate r, the average loss is $\delta^2 + r^2 \ge \delta^2$. The pre-audit law, core lower and erased-sign lower are identical in these two alternatives. Their three lower contributions therefore hold simultaneously in one worst-case total-risk bound, not for incompatible separately optimized tasks. For $M \ge 13$, $m \ge M/8$; Proposition 6.1 proves (10.2).

On an erasure, any immediately supplied virtual sign is independent of the true fair sign. The perfect target has event {virtual sign equals E_0} with probability one, whereas every source simulator has this probability at most $1 - \varepsilon$. This gives deficiency at least ε for the complete scored transcript. Reporting the known sign when available and an independent fair sign otherwise disagrees with the perfect report with probability ε. A coupling, followed by the same causal core strategy and task audit, gives the opposite inequality. The early completion requirement forbids waiting for the final audit to learn E_0. □

The joint theorem is an attained construction for one specified scored task, not a universal additive theorem for arbitrary error tolerances. Numerics adds its own relative-acceptance and readout certificates to this construction. Persistent labels, the S simulator states, scratch, program description, pulse precision, raw acquisitions and physical time remain distinct accounts.

Appendix A. Executable predictive quotients and the retained flow mechanism

Fix a declared class $\mathcal{T}$ of executable future tests, including the legal command domains, resource records and scored readouts. Suppose a countable family (T_j) determines its conditional laws. For finite visible histories h set

$$q(h) = (\mathrm{Law}(T_j \mid h))_{j\geq 1}.$$

Take Borel conditional versions on the histories under consideration. Assume the actual image $S = q(\mathsf{H})$ is Borel and admits a Borel section, legal command domains are constant on fibers, and raw report/update laws factor jointly measurably through q. These are realization hypotheses; arbitrary measurable equivalence relations need not satisfy them.

Proposition A.1 (Minimality for declared executable tests)**.** *Under these assumptions, q realizes exactly the predictive equivalence relation for $\mathcal{T}$ and carries a jointly Borel report/update kernel. Every other Borel statistic that determines these tests and admits the corresponding Borel factorization carries q measurably. For a stopped test, equality of its law follows only with the declared almost-sure termination or approximation conditions.*

Proof. Coordinate conditional laws are Borel. Their equality is precisely predictive equivalence by determination. Compose the raw kernel with a Borel section of q and push the continued history through q. Fiber constancy makes the result independent of that section and jointly Borel in state and command. Induction proves equality for finite executable prefixes. For bounded stopped readouts under almost-sure termination, truncate at time n with a distinguished surviving symbol and let $n \to \infty$; the surviving probability vanishes. For another sufficient statistic the test coordinates are constant on its fibers and the assumed factorization supplies the Borel map to q. This is minimality for the declared tests, not an unrestricted optimal-control or infinite-horizon equivalence. □

The first and second task moments in the main theorem are functions of this predictive information when its realization hypotheses hold, but they need not separate all future tests. A known finite hidden-state or quantum model has its usual conditional-state carrier; if the tests do not determine that carrier, its predictive quotient is smaller. For an unknown fixed model, the family of modelwise posteriors is a comparison object, not a single executable observer. The upper in Theorem 3.1 stores neither the exact posterior nor that family.

For comparison with classical occupation methods, we state the inherited submeasure-preserving realization separately. It is not a hidden premise of the new balanced theorem.

Proposition A.2 (Finite occupation on a Borel predictive carrier)**.** *Let S, A, Y be standard Borel with legal Borel domains, a Borel legal default policy on all states, and transition/report kernel $K(ds', dy \mid s, a)$. Let α be a prepared probability measure. Suppose a finite nonnegative legal occupation measure $\eta(ds, da)$ and a legally supported terminal measure ζ satisfy*

$$0 \leq \zeta \leq \eta K, \qquad \eta_S = \alpha + (\eta K - \zeta)_{s'}.$$

There is a stationary randomized policy with a Borel post-report stopping rule whose occupation is at most η and whose full terminal measure is exactly ζ. It terminates almost surely with finite expected count and no greater nonnegative additive costs.

Proof. Disintegrate $\eta = \xi\pi$ and choose a clipped Borel density $h = d\zeta/d(\eta K)$, zero outside the stopping domain. The continued-state kernel is $Q(s, D) = \int \pi(da \mid s) \int \mathbf{1}_D(s')(1-h)K(ds', dy \mid s, a)$. Balance is $\xi = \alpha + \xi Q$; its total mass also implies $\zeta(1) = 1$. Monotone iteration gives the actual minimal occupation $\bar{\xi} = \sum_{j\geq 0} \alpha Q^j \leq \xi$. The residual $d = \xi - \bar{\xi}$ is finite and satisfies $d = dQ$, hence $\int (1 - Q\mathbf{1})\, dd = 0$. Only rows of zero acceptance carry residual mass. With $r = d\bar{\xi}/d\xi$, the realized terminal measure is $r(s)\zeta$, and its missing mass is exactly $\int (1 - Q\mathbf{1})\, dd = 0$. Thus the terminal measure is unchanged. The actual occupation $\bar{\eta} = \bar{\xi}\pi$ is dominated by η, proving the count and cost

bounds. Finite expected count implies almost-sure commitment. Null-set choices can be completed by any measurable legal policy on the unreachable part; existence of such a completion is part of the declared Borel legal interface. □

This is a fixed-kernel Markovization statement adjacent to [7, 8, 9]. It neither gives a finite-state implementation of an arbitrary Borel policy nor merges separately chosen unknown-model policies into one program. The common finite-state implementation here comes instead from the raw moment balance and the counted rejection rule.

Appendix B. Scope and boundary examples

The center of the paper is a general sufficient-class acquired-geometry theorem, not a classification of all positive causal experiments. Its exact minimax assertion uses a fixed admissible witness with a common productive task law and an all-strategy cost identity, together with one common task-balancing program. It does not require a common primary row for all models or all strategies. If two histories in the same alleged driver state have success probabilities 1/4 and 3/4, a single exact driver row cannot represent both. The present upper still works when a proved conditional attempt bound and the task-measure balance remain valid; otherwise neither the finite-driver formula nor the new theorem is licensed.

Pointwise calibration is not necessary for these minimax laws. Static hidden signs have zero residual variance after fully revealing histories, and nearly pure quantum states can do the same. Positive current-report marks then fail on those fibers. The raw reference-plus-balance argument avoids that lower-variance premise; it does not assert that every such failed calibration can be repaired. Conversely, report-only balance and an unaffordable reference preparation are insufficient, as is a reference model whose supposedly unpredictable productive state is already leaked by a preceding report.

Action aliasing is harmless only when it preserves physical ports, costs, observable inputs and legal continuation kernels. Duplicating one physical command into two names does not increase Proposition 4.4's count. A phase-dependent dispatcher that can see an otherwise unavailable port state is an extra resource, not a relabeling. The proposition is not a universal minimal-controller characterization.

The finite-state upper is an explicit recursion: at the two physical port labels, issue the declared command; at the productive call, compute the virtual mark and finite decision; retain only the next label. It has no history tape, accurate hidden posterior, free real register, free time counter or externally randomized persistent program. Exact Borel descriptions and finite bit descriptions are distinguished by Proposition 9.2. Optimal joint program/time/workspace complexity remains unproved.

The two intervention realizations score the post-command audit. The Gaussian theorem preserves the original current-sign target and uses no intervention on it. These are different declared tasks and are never exchanged by an untyped reduction. The reference-law geometry is derived from actual preparation and its cost is included, but its existence remains an experiment-class assumption. General learning of an unknown gate lower bound, unrestricted kernel learning, arbitrary unbounded losses, nonterminating protocols, infinite-dimensional unbounded operators, and large-deviation equivalence are not conclusions of the paper. The stopped results proved here use nonnegative costs and explicit conditional hazards; they are not substitutions of a random time into a fixed-horizon theorem.

The retained predictable-geometry manuscript contains the earlier pointwise calibration and conditional-envelope route, including nonexact acquired-mass bounds. It remains valid under its stated assumptions and is preserved in full as an optional companion. Earlier curvature, continuation, adaptive refinement, causal morphism and independent realization papers likewise keep their own

hypotheses. None is used backwards to establish the central theorem here, and no historical archive is silently recertified by the present proof.

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